

ASHISH CHOUDHARY

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EDUCATION

B.Tech, Computer Science and Engineering (Data Science), VIT Vellore

Expected 2028

CGPA: 9.07 | Relevant Coursework: Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Architecture & Organization, Theory of Computation, Software Engineering

12th Grade, FIITJEE Jr. College (94.2%)

2022 - 2024

10th Grade, Kennedy High the Global School (93%)

2010 - 2022

EXPERIENCE

AI Engineering Intern, Cyient, Hyderabad

June 2026 - July 2026

Tech Stack: Python, Azure, OpenCV, NumPy, Pandas, OCR, Object-Oriented Design

- Designed and implemented rule-based classification algorithms in Python to automate remediation workflows for validation and verification of qualification procedures, improving process efficiency and reducing manual effort.
- Built an end-to-end automated CoC and CoA annotation system using object-oriented design principles in Python, applying OpenCV contour detection and probabilistic OCR via Document Intelligence Azure text extraction to cut manual processing time by 2-3 hours per certificate.
- Engineered a scalable data pipeline in Python to automate periodic reporting, consolidating large-scale timesheet data into manager-wise KPI summaries across offshore and onsite locations, improving data reliability and turnaround time. Collaborated with cross-functional teams in an Agile environment using Git for version control.

PROJECTS

C.O.T.E.ai: Concept Oriented Training and Execution

[GitHub](#)

Tech Stack: React.js, FastAPI, Python, LangChain, ChromaDB, Gemini API, REST APIs, RAG, OCR

- Architected and developed a full-stack, AI-powered Learning Management System with a React.js front end and a FastAPI REST backend, extending core functionality beyond platforms like Google Classroom.
- Designed a hybrid Retrieval-Augmented Generation (RAG) pipeline with multi-query retrieval and vector-based search using ChromaDB, increasing context accuracy by 25%.
- Automated multilingual flashcard generation using LangChain, reducing manual study-preparation time by 60% through efficient information retrieval.
- Optimized backend inference through recursive chunking and API call reduction, cutting API token costs by 30% and improving system scalability.

myshell: POSIX-Compliant Unix Shell

[GitHub](#)

Tech Stack: C++17, POSIX System Calls, CMake, Linux/UNIX

- Built a POSIX-compliant shell in C++17 implementing the fork-exec-wait process model, I/O redirection, and multi-stage pipelines using low-level syscalls (fork, execvp, pipe, dup2, waitpid).
- Designed an RAII wrapper around POSIX file descriptors for exception- and error-safe resource cleanup across pipe/redirect operations, eliminating manual close() bookkeeping.
- Implemented background job control and signal handling (SIGINT, SIGCHLD) for asynchronous process execution without blocking the shell's main loop.
- Wrote a recursive-descent parser to tokenize shell input into commands, pipelines, and redirection targets, decoupled from execution logic.

FrameByFrame: Full-Stack Visual Trivia Platform

[GitHub](#) | [Live](#)

Tech Stack: Python, FastAPI, SQLAlchemy, PostgreSQL, React, TypeScript, Docker, Vercel, Render

- Architected and shipped a full-stack web trivia game end-to-end: a layered FastAPI backend (routes → services → repositories) with SQLAlchemy + Alembic-managed PostgreSQL, deployed via Docker across Vercel and Render with automated production deploys.
- Engineered server-authoritative, race-condition-safe gameplay logic using database-clock timestamps, row-level locking, and idempotent session creation to guarantee consistent scoring under concurrent requests.
- Built a Decimal-precision scoring engine and HMAC-signed guest authentication, backed by 14+ automated test suites (pytest, Vitest) covering gameplay logic, scoring, and content management.

TECHNICAL SKILLS

Languages: Python, Java, C++, C, JavaScript, TypeScript

Core CS Fundamentals: Object-Oriented Programming (OOP), Computer Architecture & Organization, Database Management Systems (DBMS), Operating Systems, Theory of Computation, Software Engineering

Systems & Distributed Computing: Concurrency, Multi-threading, REST API Design, Microservices, Docker, AWS

AI / ML: PyTorch, Scikit-Learn, LangChain, Machine Learning, OpenCV, RAG Pipelines, LLMs, NLP

Data & Tools: NumPy, Pandas, PostgreSQL, SQL, Vector Databases (ChromaDB, FAISS), Hugging Face, Sentence-Transformers

Web & Backend Frameworks: React.js, FastAPI, Node.js, Tailwind CSS

Developer Tools & Practices: Git, GitHub, Postman, Agile Development, Debugging, Unit Testing

Cloud Computing: Azure, AWS, Vercel, Render

ACHIEVEMENTS

- Placed 11th nationally in the KRUU Grasp national hackathon among competitive software engineering teams.
- Finalist, AI for Bharat hackathon by AWS, for the C.O.T.E.ai application: recognized for technical architecture and scalability.
- 99th percentile in Mathematics, JEE Mains: strong foundation in quantitative problem solving and algorithmic thinking.